

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	1	2	3	1	2	1	1	2	3	2	4	3	1	3	1	2	2	3	1	3	4	3	4	3	2	2	2	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	3	2	2	2	4	2	1	3	3	2	2	4	3	1	4	2	2	4	2	1	4	1	4	3	1	4	3	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	4	4	1	4	4	4	2	4	3	1	1	2	4	3	2	2	1	1	1	2	4	2	1	2	4	4	1	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	2	2	3	4	3	4	2	2	2	4	2	3	4	4	2	1	3	2	4	3	4	1	1	3	3	2	1	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	4	4	1	3	1	2	1	4	2	1	1	4	2	3	3	3	3	2	4	4	3	3	1	3	2	1	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	4	4	2	2	4	2	4	4	2	2	2	1	3	1	4	3	2	2	1	2	3	2	1	2	4	4	3	3	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	4	3	4	3	2	3	3	1	3	1	3	2	4	1	4	4	4	2	2	2										

HINT - SHEET

SUBJECT : PHYSICS

SECTION - A

1. Ans (4)

$$\text{emf } e = \frac{\phi_f - \phi_i}{\Delta t} \quad \phi_i = NBA$$

$$e = - \frac{(-NBA - NBA)}{t} = \frac{2NBA}{t}$$

$$= \frac{2 \times 35 \times 10^2 \times 10^{-4} \times \pi \times 0.5 \times 0.5 \times 10^{-6}}{0.1}$$

$$= 5.5 \mu V$$

2. Ans (1)

Given $e = E_0 \sin \omega t$

$$i = I_0 \sin (\omega t + \phi)$$

So phase difference = ϕ

power factor = $\cos \phi$

By formula

$$\text{Average power } (P_{av}) = V_{rms} I_{rms} \cos \phi$$

$$= \frac{E_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \cos \phi = \frac{E_0 I_0}{2} \cos \phi$$

3. **Ans (2)**

For points inside the wire

$$B = \frac{\mu_0 I r}{2\pi R^2} \quad (r \leq R)$$

For points outside the wire

$$B = \frac{\mu_0 I}{2\pi r} \quad (r \geq R)$$

according to the question

$$\frac{B}{B'} = \frac{\frac{\mu_0 I (a/2)}{2\pi a^2}}{\frac{\mu_0 I}{2\pi (4a)}} = 2 : 1$$

4. **Ans (3)**

$$i = \frac{18}{4+2} = 3A$$

$$P = i^2 R = 3^2(4) = 36W$$

5. **Ans (1)**

$$4 = \frac{S(6)}{S+6}$$

$$\Rightarrow S = 12\Omega$$

6. **Ans (2)**

The current through $N = 3A$

$$\Delta V \text{ across } 1\Omega = iR$$

$$= 3(1) = 3V$$

7. **Ans (1)**

$$R = (2\pi r) \times 24$$

$$R_{eq} = \frac{\frac{R}{2} \frac{R}{2}}{\frac{R}{2} + \frac{R}{2}} = \frac{R}{4} = 1.2\pi\Omega$$

8. **Ans (1)**

$$q_i = cV$$

$$V_C = \frac{V}{2}$$

$$q_f = \frac{cV}{2} = \frac{q_i}{2}$$

9. **Ans (2)**

$$x = A \sin \omega t$$

$$v = A\omega \cos \omega t$$

$$a = -A\omega^2 \sin \omega t = -\omega^2 x$$

10. **Ans (3)**

Efficiency of carnot engine

$$\eta = \left(1 - \frac{T_2}{T_1}\right)$$

It depends on temperature of source & sink.

11. **Ans (2)**

$$(V_{rms})_{Ne} = (V_{rms})_{NH_3}$$

$$\sqrt{\frac{3RT}{20}} = \sqrt{\frac{3R(290)}{17}}$$

$$T = \frac{20 \times 290}{17} = 341.1K$$

12. **Ans (4)**

$$P_{mix} = \frac{(\mu_1 + \mu_2)RT_{mix}}{V_{mix}} = \frac{(0.1 + 0.2) \times 8.3 \times 200}{100 \times 10^{-6}} = 4.98 \times 10^6 \text{ Pascal}$$

13. **Ans (3)**

A spaceship is a vehicle that travels through space. When spaceship enters earth's atmosphere, it experiences air friction due to which it will catch fire, if proper precaution will not be taken. Also, temperature of upper atmosphere is low,

14. **Ans (1)**

$$V_1 = \frac{m - e(2m)}{m + 2m} \quad V = 0 \Rightarrow e = \frac{1}{2}$$

15. **Ans (3)**

Tension is maximum at lowest point.

16. **Ans (1)**

For convex lens, it forms a real image

$$|u| + |v| \geq 4f,$$

$$\text{for } |u| = 2f, |v| = 2f$$

$$\text{So } |u| + |v| = 4f.$$

17. Ans (2)

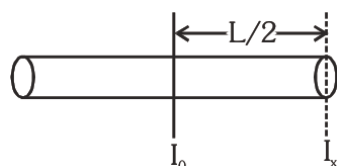
$$\frac{1}{2} Kx^2 = \frac{1}{2} m \left[u^2 - \left(\frac{u}{2} \right)^2 \right]$$

$$\Rightarrow Kx^2 = m \frac{3u^2}{4}$$

$$\Rightarrow x = \sqrt{\frac{3mu^2}{4k}} = \frac{u}{2} \cdot \sqrt{\frac{3m}{k}}$$

18. Ans (2)

Using parallel axis theorem



$$I_x = I_0 + \frac{ML^2}{4}$$

19. Ans (3)

$$a_1 = \frac{g \sin \theta}{1 + \frac{2}{3}} = \frac{3}{5} g \sin \theta$$

$$a_2 = g \sin \theta$$

$$\frac{a_1}{a_2} = \frac{3}{5}$$

20. Ans (1)

$$I_g = \frac{F}{m} = \frac{3}{60 \times 10^{-3}} = 50 \text{ N/kg}$$

21. Ans (3)

$$U = \frac{1}{2} (\text{force})(\text{elongation})$$

$$= \frac{1}{2} (Mg)\ell = \frac{1}{2} Mg\ell$$

22. Ans (4)

$$AB \sin \theta = AB \cos \theta$$

$$\tan \theta = 1$$

$$\theta = 45^\circ$$

23. Ans (3)

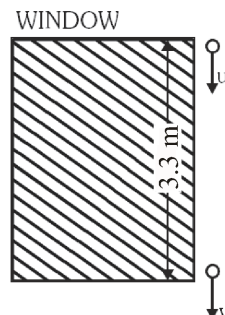
$$\frac{MV^2}{T} = \frac{ML^2T^{-2}}{T} = ML^2T^{-3} = \text{Power}$$

24. Ans (4)

$$g = 4\pi^2 \frac{L}{T^2}$$

$$\therefore \frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T}$$

25. Ans (3)



$$\text{By using } s = ut + \frac{1}{2} at^2$$

$$3.3 = u(0.2) + \frac{1}{2} (10) (0.2)^2$$

$$\Rightarrow 33 = 2u + 2$$

$$\Rightarrow u = \frac{31}{2} = 15.5 \text{ ms}^{-1}$$

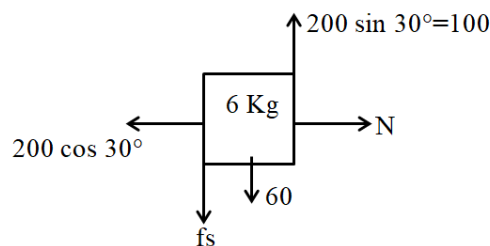
26. Ans (2)

$$t = \sqrt{\frac{2h}{a_{\text{rel}}}}$$

$$\text{In second case } a_{\text{rel}} = a + g$$

$$\text{Hence } t_1 > t_2$$

27. Ans (2)



$$N = 200 \cos 30^\circ$$

$$= 100 \sqrt{3}$$

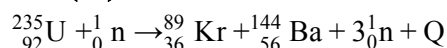
$$= 173 \text{ N}$$

$$f_{\text{SL}} = \mu_s N = \frac{1}{4} \times 173$$

$$f_{\text{SL}} = 43.25$$

$$f_{\text{req}} = 100 - 60 = 40 \text{ N} < f_{\text{SL}}$$

28. Ans (2)



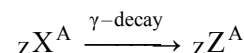
29. Ans (3)

$$E = mc^2$$

$$= 0.5 \times 10^{-3} \times 9 \times 10^{16}$$

$$= 4.5 \times 10^{13} \text{ J}$$

30. Ans (2)



Due to gamma emission, there is no change in mass number and atomic number.

31. Ans (4)

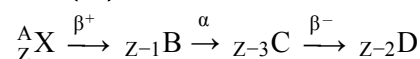
$\text{X}^{240} \rightarrow \text{Y}^{120} + \text{Z}^{120}$ given binding energy per nucleon of X, Y & Z are 7.6 MeV, 8.5 MeV & 8.5 MeV respectively.

Gain in binding energy is :-

Q = Binding Energy of products - Binding energy of reactants

$$= (120 \times 8.5 \times 2) - (240 \times 7.6) \text{ MeV} = 216 \text{ MeV}$$

32. Ans (3)



β^+ decreases atomic number by 1

α decreases atomic number by 2

β^- increases atomic number by 1

33. Ans (2)

Path difference for nth max = $n\lambda = 6\lambda$

34. Ans (2)

For P type

Holes are majority & trivalent atoms are the dopants.

35. Ans (2)

$$\overline{Y} = \overline{A + B}$$

$$\overline{Y} = \overline{A \cdot B} = \overline{A} \cdot \overline{B} = A \cdot B = \text{AND gate}$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

SECTION - B

36. Ans (4)

$$I = \frac{E}{At} \Rightarrow E = IAt = \frac{20}{10^{-4}} \times 20 \times 10^{-4} \times 60$$

$$= 24 \times 10^3 \text{ J}$$

37. Ans (2)

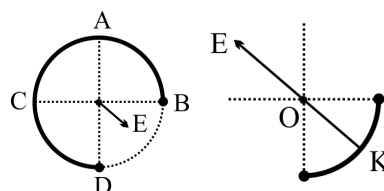
$$\text{Since } \ell = 2\pi R = n(2\pi r) \Rightarrow r = \frac{R}{n}$$

$$\text{For two turn } B = \frac{2\mu_0 I}{2R/2} \text{ and } B = 4 \left(\frac{\mu_0 I}{2R} \right)$$

$$\text{For } n \text{ turn } \Rightarrow B' = n^2 \left(\frac{\mu_0 I}{2R} \right) \Rightarrow B' = \frac{n^2 B}{4}$$

38. Ans (1)

Net field at O due to whole ring = 0



39. Ans (3)

$$R = \frac{V}{I} = \tan 53^\circ = \frac{4}{3} \Omega ; r = 10^{-3} \text{ m}$$

$$\frac{4}{3} = \frac{\rho \ell}{A} \Rightarrow \rho = \frac{4}{3} \frac{A}{\ell} = \frac{4}{3} \frac{(3.14 \times 10^{-6})}{(0.5)}$$

$$\rho \approx 8 \times 10^{-6} \Omega\text{-m}$$

40. Ans (3)

Root mean square speed,

$$V_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3RT}{m}}$$

$$\text{Pressure exerted by gas } P = \frac{1}{3}(\text{nm})\bar{v}^2$$

$$\Rightarrow P = \frac{1}{3}\rho\bar{v}^2$$

$$\text{Average kinetic energy K.E.} = \frac{3}{2}KT$$

Total internal energy of 1 mole of a

$$\text{monoatomic} \Rightarrow \frac{3}{2}RT$$

41. Ans (2)

$$P^2V = \text{Constant}$$

$$\left(\frac{nRT}{V}\right)^2 \times V = \text{Constant}$$

$$\frac{T^2}{V} = \text{Constant}$$

$$\therefore T_1 < T_2 \Rightarrow V_1 < V_2$$

42. Ans (2)

Case-I : frequency can be 496 or 504 Hz

Case-II : frequency can be

$$\frac{1510-2}{3}, \frac{1510+2}{3}$$

$$: 502.6 ; 504 \text{ Hz}$$

504 Hz satisfy both cases

43. Ans (4)

$$P_i = P_f \Rightarrow (1+3) \left(20\hat{i} + 25\hat{j} - 12\hat{k} \right)$$

$$= 1 \left(100\hat{i} + 35\hat{j} + 8\hat{k} \right) + 3\vec{v}$$

$$\Rightarrow \vec{v} = \frac{-20\hat{i} + 65\hat{j} - 56\hat{k}}{3} \text{ m/s}$$

44. Ans (3)

$$W = 3F_0 \times 2d + \frac{\pi d(2F_0)}{4}$$

$$= F_0 d \left(6 + \frac{\pi}{2} \right)$$

45. Ans (1)

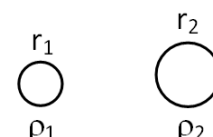
$$100 w_d = (50 \times 1^2) (2/1) \Rightarrow w_d = 1 \text{ rad/s}$$

$$w_m = \frac{v}{r} = \frac{2}{1} = 2 \text{ rad/s}$$

$$w_{md} = w_m - w_d = 2 - (-1) = 3$$

$$\Rightarrow \frac{2\pi}{T_{md}} = 3 \Rightarrow T_{md} = \frac{2\pi}{3} \text{ s}$$

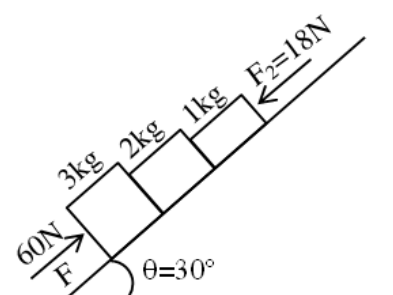
46. Ans (4)



$$v_T = \frac{2r^2(\sigma - \rho)g}{9\eta} \Rightarrow \frac{v_1}{v_2} = \left(\frac{r_1}{r_2} \right)^2 \frac{(\sigma_1 - \rho)}{(\sigma_2 - \rho)}$$

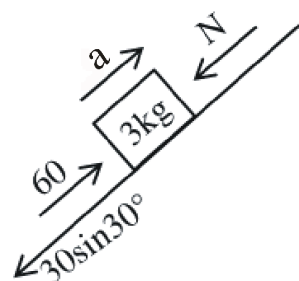
$$= \left(\frac{1}{2} \right)^2 \left(\frac{8\rho_2 - 0.1\rho_2}{\rho_2 - 0.1\rho_2} \right) = \frac{79}{36}$$

47. Ans (2)



$$a = \frac{\text{Net force}}{\text{Total mass}} = \frac{60 - (18 + 60 \sin 30^\circ)}{6} = 2 \text{ ms}^{-2}$$

FBD of 1kg



$$60 - 15 - N = 3 \times 2$$

$$45 - N = 6$$

$$N = 39 \text{ N}$$

48. Ans (2)

If no shunt

$$I = 9 \times 60 \mu\text{A} = 540 \times 10^{-6} \text{ A}$$

$$\frac{6}{11000 + G} = 540 \times 10^{-6}$$

$$G = \frac{1000}{9}$$

$$G = \frac{RS}{R - S} \Rightarrow \frac{1000}{9} = \frac{11000 S}{11000 - S}$$

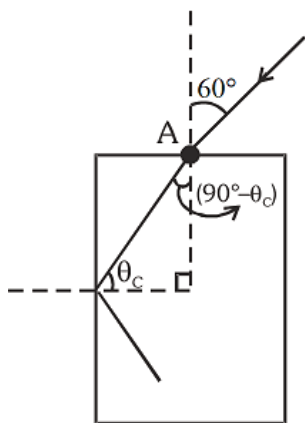
$$S = 110 \Omega$$

49. Ans (4)

at A Snell's law

$$1 \times \sin 60^\circ = \mu \sin(90^\circ - \theta_C)$$

$$\frac{1}{\sqrt{2}} = \mu \cos \theta_C \quad \dots(1)$$



For second surface

$$\frac{1}{\mu} = \sin \theta_C \quad \dots(2)$$

According to (1) and (2)

$$\sin^2 \theta_C + \cos^2 \theta_C = 1$$

$$\cos \theta_C = \sqrt{1 - \sin^2 \theta_C} \text{ put in (1)}$$

$$\frac{\sqrt{3}}{2} = \mu \times \sqrt{1 - \frac{1}{\mu^2}} \Rightarrow \frac{\sqrt{3}}{2} = \sqrt{\mu^2 - 1}$$

$$\sqrt{\frac{3}{4} + 1} = \mu \Rightarrow \mu = \frac{\sqrt{7}}{2} = \sqrt{\frac{7}{4}}$$

50. Ans (2)

$$KE_1 = \frac{hc}{\lambda} - \phi$$

$$KE_2 = \frac{hc}{\lambda/2} - \phi = \frac{2hc}{\lambda} - \phi$$

$$KE_2 = 3KE_1$$

$$\Rightarrow \frac{2hc}{\lambda} - \phi = 3 \left(\frac{hc}{\lambda} - \phi \right)$$

$$\Rightarrow 2\phi = \frac{hc}{\lambda}$$

$$\Rightarrow \phi = \frac{hc}{2\lambda}$$

SUBJECT : CHEMISTRY

SECTION-A

51. Ans (1)

NCERT XIIth Pg. # 318 (Example 10.7)

52. Ans (4)

NCERT XIIth Pg. # 347

53. Ans (1)

NCERT XIIth Pg. # 348

54. Ans (4)

NCERT XIIth Pg. # 382

55. Ans (3)

NCERT Pg. #421,422

56. Ans (1)

NCERT Pg. # 399

57. Ans (4)

NCERT Pg. #442 table 14.3

59. Ans (3)

Zn and Mn has equal metallic radius

63. Ans (4)

NCERT Pg. # 319

64. **Ans (4)**

NCERT Pg. # 324

69. **Ans (2)**

NCERT XI PART-1 Pg.#18

70. **Ans (4)**

NCERT XIth Part-1 Pg.#161

71. **Ans (3)**

NCERT XIth Part-1 Pg. # 51, 70, 46, 63

72. **Ans (1)**

By law of equivalence

(No. of g - eq) Na₂SO₃ = (No. of g - eq) KMnO₄

$$n_1 \times n_{f_1} = n_2 \times n_{f_2}$$

$$n_1 \times 2 = 3 \times 5 \Rightarrow n_1 = \frac{15}{2} = 7.5$$

73. **Ans (1)**

NCERT XIth Part-I Pg. # 41,50

74. **Ans (2)**

NCERT, XIth, Part - I Pg.#187, 208, 211, 212

75. **Ans (4)**

$$(NV)_{\text{Strong Acid}} = 20 \times 0.25 = 5$$

$$(NV)_{\text{Strong base}} = 30 \times 0.2 = 6$$

$$\because (NV)_{\text{SB}} > (NV)_{\text{SA}}$$

⇓

solution will be basic

$$[\text{OH}^-] = \frac{(NV)_{\text{SB}} - (NV)_{\text{SA}}}{V_T}$$

$$= \frac{6 - 5}{50} = \frac{1}{50} = 0.02 \text{ N}$$

77. **Ans (2)**

$$i = \frac{\text{Theoretical M. w.}}{\text{Observed M. w.}} = \frac{170}{140} = 1.214$$

$$i = 1 + (n - 1) \alpha$$

$$1.214 = 1 + (2 - 1) \alpha \Rightarrow \alpha = 0.214 = 21.4 \%$$

78. **Ans (2)**

For uncatalysed reaction $k_1 = Ae^{-E_a/RT}$

For uncatalysed reaction $k_2 = Ae^{-\frac{(E_a - 8314)}{RT}}$

$$\frac{k_2}{k_1} = e^{\frac{E_a}{RT} - \left(\frac{E_a - 8314}{RT}\right)}$$

$$= e^{\frac{8314}{RT}} = e^2 = 7.38$$

79. **Ans (1)**

Rate law = $r = k[A]^1[B]^1$

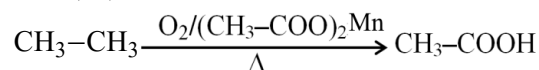
its a second order reaction so unit of rate

constant will be $\text{L mol}^{-1}\text{sec}^{-1}$

84. **Ans (2)**

if R.S. are not identical then single bond acquires more single bond character and double bond character.

85. **Ans (1)**



SECTION-B

86. **Ans (2)**

NCERT XIIth Pg. # 345

87. **Ans (4)**

Glucose and Fructose both gives tollen's test.

91. **Ans (4)**

NCERT Pg. # 319

92. **Ans (3)**

NCERT Pg.#123

93. Ans (2)

NCERT XIth Part - 1 Pg. # 220

94. Ans (2)

NCERT XIth Part-I Pg.#175

96. Ans (4)

$$P_T = P_A^{\circ}X_A + P_B^{\circ}X_B$$

$$P_T = P_A^{\circ}(1 - X_B) + P_B^{\circ}X_B$$

$$P_T = P_A^{\circ} - P_A^{\circ}X_B + P_B^{\circ}X_B$$

$$P_T = P_A^{\circ} + X_B (P_B^{\circ} - P_A^{\circ})$$

$$600 = 450 + X_B (700 - 450)$$

$$150 = X_B \times (250)$$

$$X_B = \frac{150}{250} = 0.6$$

Now,

$$P_B^{\circ}X_B = Y_B \cdot P_T$$

$$700 \times 0.6 = Y_B \times 600$$

$$\Rightarrow Y_B = 0.7$$

97. Ans (3)

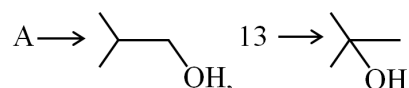
$$k = \frac{1}{t} \ln \frac{[A_0]}{[At]} \Rightarrow t_{90\%} = \frac{1}{k} \ln \frac{(A_0)}{\left(\frac{A_0}{10}\right)} = \frac{\ln 10}{k}$$

$$(1) t_{90\%} = \frac{2.303}{k}$$

$$(2) t_{1/2} = \frac{2.303 \times \log 2}{k}$$

$$\frac{(1)}{(2)} t_{90\%} = \frac{t_{1/2}}{\log 2} = 3.3 t_{1/2}$$

100. Ans (2)



SUBJECT : BOTANY

SECTION - A

101. Ans (2)

NCERT XII, Pg # 75

102. Ans (4)

NCERT XII, Pg # 112

103. Ans (2)

NCERT XII, Pg # 73

104. Ans (3)

NCERT XII, Pg # 98

105. Ans (4)

NCERT XII, Pg # 89

106. Ans (4)

NCERT XII, Pg # 122

107. Ans (2)

NCERT XI, Pg. # 38

108. Ans (1)

NCERT XI, Pg. # 33

109. Ans (3)

NCERT XI, Pg # 32

110. Ans (2)

NCERT XII, Pg. # 28

111. Ans (4)

NCERT-XIIth, Pg.# 111

112. Ans (3)

NCERT XI, Pg # 84, 88

113. Ans (4)

NCERT XI, Pg # 89, 90, 91, 92, 93

114. Ans (1)

NCERT XI, Pg # 75

115. Ans (1)

NCERT XI, Pg # 213

116. Ans (3)

NCERT XI Pg # 229, 231, 233

117. **Ans (3)**
NCERT XI Pg # 246
118. **Ans (2)**
NCERT XII (Pg = 235)
119. **Ans (1)**
NCERT XII (Pg - 233)
120. **Ans (2)**
NCERT, Pg. # 84
121. **Ans (2)**
NCERT XII, Pg. # 92
122. **Ans (1)**
NCERT, Pg. # 94, 95
123. **Ans (4)**
NCERT XI Pg. No. # 75
124. **Ans (4)**
NCERT XI Pg. # 23, 24
125. **Ans (1)**
NCERT XI, Pg. # 14
126. **Ans (3)**
NCERT-XI, Pg.# 30
127. **Ans (1)**
NCERT-XI, Pg.# 247
128. **Ans (2)**
NCERT XI, Pg. # 98
129. **Ans (1)**
NCERT XII, Pg. # 22
130. **Ans (4)**
NCERT XII Pg. No. # 213
131. **Ans (2)**
NCERT XII, Pg # 207

132. **Ans (1)**
NCERT XII, Pg. # 249
133. **Ans (1)**
NCERT-XII Pg. # 255
134. **Ans (4)**
NCERT XI, Pg.#231
135. **Ans (2)**
NCERT XII, Pg. # 81

SECTION - B

136. **Ans (3)**
NCERT XII, Pg # 110
137. **Ans (3)**
NCERT XI, Pg.# 35
138. **Ans (3)**
NCERT XI, Pg # 31
139. **Ans (3)**
NCERT XI, Pg. # 19, 24
140. **Ans (2)**
NCERT XI, Pg. # 11
141. **Ans (4)**
NCERT XI, Pg # 75, 78, 79
142. **Ans (4)**
NCERT XI Pg # 219
143. **Ans (3)**
NCERT XI Pg # 250
144. **Ans (3)**
NCERT XII Pg. # 242
145. **Ans (1)**
NCERT XII Page No. 242
146. **Ans (3)**
NCERT XII, Pg. # 221

147. **Ans (2)**
NCERT XII, Pg. # 224, 225, 226

148. **Ans (1)**
NCERT XI, Pg. # 212

149. **Ans (3)**
NCERT XII, Pg. # 152, 153

150. **Ans (4)**
NCERT, Pg. # 98, 99

SUBJECT : ZOOLOGY**SECTION - A**

151. **Ans (1)**
NCERT-XII, Pg. # 58, 59

152. **Ans (4)**
NCERT, Pg. # 339

153. **Ans (4)**
NCERT, Pg. # 105

154. **Ans (2)**
NCERT, Pg. # 50, 51, 52, 53

155. **Ans (2)**
NCERT, Pg. # 274

156. **Ans (4)**
NCERT, Pg. # 281

157. **Ans (2)**
NCERT, Pg. # 272

159. **Ans (4)**
NCERT, Pg. # 49, 50, 51, 54

160. **Ans (2)**
NCERT, Pg. # 49, 51, 52, 54

161. **Ans (2)**
NCERT, Pg. # 307, 308

162. **Ans (2)**
NCERT Pg.#201

163. **Ans (1)**
NCERT Pg.#202

164. **Ans (3)**
NCERT Pg.#198

165. **Ans (1)**
NCERT Pg.#199

166. **Ans (4)**
NCERT Pg.#196

167. **Ans (3)**
NCERT Pg.#211

168. **Ans (2)**
NCERT Pg.#202

169. **Ans (2)**
NCERT Pg.#212

170. **Ans (1)**
NCERT Pg.#212

171. **Ans (2)**
NCERT XIth Pg. No. # 312

172. **Ans (3)**
NCERT XII Pg. No.# 60

173. **Ans (2)**
NCERT XII Pg. No. # 51

174. **Ans (1)**
NCERT Pg. No. # 63

175. **Ans (2)**
NCERT Pg. No. # 60-61

176. **Ans (4)**
NCERT - XII Pg. No. # 54 - 55

177. **Ans (4)**
NCERT XI Pg # 310-311

178. **Ans (3)**
NCERT XII Pg. No. # 53

179. **Ans (3)**
NCERT XII Pg. No. # 211

180. **Ans (3)**
NCERT - XII Pg. No. # 147 - 149

181. **Ans (4)**
NCERT Pg.# 146

182. **Ans (3)**
NCERT Pg.# 138

183. **Ans (4)**
NCERT-XII, Pg.# 201

184. **Ans (3)**
NCERT Pg.# 168

185. **Ans (2)**
NCERT Pg. # 108

SECTION - B

186. **Ans (3)**
NCERT, Pg. # 108

187. **Ans (3)**
NCERT Pg. # 106

188. **Ans (1)**
NCERT-XII, Pg.# 213

189. **Ans (3)**
NCERT Pg. No. # 22

190. **Ans (1)**
NCERT, Pg. # 108, 109

191. **Ans (3)**
Module, Pg. # 14, 15

192. **Ans (2)**
NCERT (XI) Pg. # 279; para-2

193. **Ans (4)**
NCERT Page no. 332

194. **Ans (1)**
NCERT–XI, Pg. # 50

195. **Ans (4)**
NCERT (XI) Pg # 274

196. **Ans (4)**
NCERT(XIIth) Pg # 152

197. **Ans (4)**
During oogenesis M-I division start before birth & complete during reproductive age means during each menstrual cycle just prior to ovulation.
अण्डजनन के दौरान प्रथम अर्द्धसूत्री विभाजन जन्म से पूर्व शुरू होता है और अन्त प्रत्येक मासिक चक्र के दौरान अण्डोत्सर्ग से ठीक पूर्व होता है।

198. **Ans (2)**
NCERT-XI, Pg No.102, IInd para

199. **Ans (2)**
NCERT XI Pg.# 143

200. **Ans (2)**
NCERT-XI, Page#146